

LESSON 3.7

SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS – PART 3



LESSON INTRODUCTION

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

LEARNING TARGETS

- I can solve problems with three operations involving any of the four operations with rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How can you determine which form of rational number should be used when solving real-world problems with fractions and decimals?
- How do you solve multi-step real-world problems involving any of the four operations with rational numbers?

LESSON NARRATIVE

Lesson 7 extends students' comprehension of real-world problems to analyze the work of others within that context. 3.7.2 Activity drives learning outcomes in this lesson, with students actively engaging with each other and with real-world content by analyzing the written work of students provided in the Workbook. Students are not only identifying which operations are used in each context, but also evaluating the approaches, algorithms, and calculations of their peers.

COMPONENT SUMMARY

3.7.1 WARM-UP (~5 MIN.)

Label the order of operations in the correct sequence

3.7.2 ACTIVITY (25 MIN.)

Analyze examples of student work using operations and rational numbers

3.7.3 COLLABORATION (10 MIN.)

Fill in missing information for a real-life example using operations and rational numbers

3.7.4 WRAP-UP (~5 MIN.)

Self-reflection over a positive aspect, progress made, and a confusing part from the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Order of operations

MATERIALS

- (Optional) Calculator

ADDITIONAL PREPARATION

- An anchor chart showing the processes to convert equivalent rational numbers may be useful.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may use the incorrect operation to find the fraction of a number.
- Students may use the incorrect place value to round decimals that represent money, like using .5 for five cents or .05 for 50 cents.

THINGS TO CONSIDER

- 3.7.2 Activity includes performing operations with several different forms of rational numbers. Consider posting or reviewing the processes to convert to equivalent forms of rational numbers.
- For question 4 in the 3.7.2 Activity, students may need support in understanding why they need to add the ratios first to find the number of ounces in one batch of dye before finding how much of each dye they need for 20 oz.
- Consider not allowing students to use a calculator to build students fluency with operations using rational numbers.
- Consider creating an observational chart to use as you circulate. Determine what aspects of the learning target, the benchmark, and MTR standards you would like to observe.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

3.7.2 Activity provides an opportunity for students to engage in a Think-Pair-Share routine as they complete Error Analyses for four student work samples. Encourage students to consider each display of student work, share their analysis with a partner, and revise or build on their responses before putting their answers in writing. This encourages mathematical discussion and deepens conceptual understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Students are actively engaging in analyzing the work of others throughout this lesson, particularly in working through 3.7.2 Activity. Activities like this facilitate a “culture in which students ask questions of the teacher and their peers, and an error is an opportunity for learning,” as stated in MA.K12.MTR.4.1 clarifications. Encourage and celebrate questions and students who change or adjust their thinking based on the questions or reasoning of others.

DIFFERENTIATE INSTRUCTION

Task Cards of the student examples could be provided for the 3.7.2 Activity.

- Students can engage with each task card at stations to incorporate movement and place when solving problems. For example, you could decorate each station to anchor students in the context (i.e., task card 1 could be the Six Flags station).
- Instead of rotations with different task cards, students could be assigned numbered groups to work through a set of Task Cards to then find differing partners or pairs to either compare and verify answers, or to teach as a small-group instructor to their peers.
- For students who struggle with calculations consider incorporation a calculator station. Each pair may be given a “ticket” which they can trade in to use the calculator.

3.7.1 WARM-UP: BELL WORK



3.7.1 Warm-Up: Bell Work

1. Number each operation using numbers 1-4 to reflect the correct order of the order of operations.

4	Perform addition and subtraction in order from left to right. (→)
1	Perform operations inside grouping symbols.
2	Evaluate expressions with exponents.
3	Perform multiplication and division in order from left to right. (→)

 3.7.2 Activity: Error Analysis

With your partner, answer each question below.

1. Diana was given the following problem to solve.
- Eugene has \$40.00 to spend at a theme park gift shop. He receives a discount of $\frac{1}{5}$ off the price of an item for being an annual passholder. He wants to purchase a collectible passholder pin that has a retail price of \$19.95. How much will he have left after purchasing the pin?

Diana solved the problem with the steps shown.

Diana's Work

$$19.95 \cdot (-\frac{1}{5})$$
$$\frac{19.95}{5} \cdot (-\frac{1}{5})$$
$$\frac{19.95}{5}$$
$$\begin{array}{r} 3.99 \\ -15 \\ \hline 49 \\ -45 \\ \hline 45 \\ -45 \\ \hline 0 \end{array}$$

$$\$40.00 - \$3.99$$
$$\begin{array}{r} 399 \\ -399 \\ \hline 3601 \end{array}$$

Eugene had \$36.01 left.

a. Identify what Diana did correctly and incorrectly.

Correct	Incorrect
Diana determined that the discount from being an annual passholder was \$3.99.	Instead of taking the discount from the cost of the collectable passholder pin, Diana subtracted the discount from the total amount Eugene had to spend.

- b. How much money will Eugene have left over after buying the pin? Show your work.
- \$24.04**

3.7.2 ACTIVITY: ERROR ANALYSIS

TEACHER MOVES

MA.K12.MTR.4.1: Error Analysis

Continue encouraging students to analyze the work of others throughout this activity. Consider what things you look for as a teacher and prompt students to utilize the same strategies for analysis.

DIFFERENTIATE INSTRUCTION

Task Cards

Consider setting up the student work examples as Task Cards for students to rotate in groups or stations. It may be helpful to group students based on varying mastery levels to ensure groups do not get stuck and students can collaborate effectively to solve each problem.

TEACHER MOVES

In order to assist students with analyzing student work, urge them to number and write out the steps from each student example. This provides opportunities for students to use specific mathematical language to communicate correct and incorrect steps and calculations.

TEACHER MOVES

Think-Pair-Share

Consider engaging students in a Think-Pair-Share routine as they identify what each student did correctly and incorrectly for questions 1-4. Position students to analyze and build on each other's responses before putting their answers in writing.

3.7.2 ACTIVITY: ERROR ANALYSIS (CONTINUED)

TEACHER MOVES

Students may struggle to represent Diego's starting elevation because it is below sea level and will therefore have trouble finding the error in Jeffrey's work. Ask questions to further scaffold this problem:

- What does it mean if the starting elevation is -262 feet?
- How can you represent a change in elevation when the starting point is below sea level and the end point is above sea level?

2. Jeffrey was given the following problem to solve.

Diego is planning to hike the very difficult Shorty's Well Route to Telescope Peak, which begins at an elevation of -262 feet and ends at an elevation of 11,043 feet. He wants to hike $\frac{3}{5}$ of the change in elevation of the trail on the first day of his hike. If Diego sticks to his plan, at what elevation will he start on the second day?

Jeffrey solved the problem with the steps shown.

Jeffrey's Work

$$\begin{array}{r}
 11043 - 262 = \\
 \underline{10781} \text{ feet} \\
 10781 \cdot \frac{3}{5} = \frac{32343}{5}
 \end{array}$$

Diego plans to be at 6468.6 feet to start Day 2.

- a. Identify what Jeffrey did correctly and incorrectly.

Correct	Incorrect
Jeffrey understood that he needed to multiply the change by $\frac{3}{5}$.	Jeffrey first needs to find out how much elevation the trail changes. He should have subtracted the highest elevation from the lowest to find this. His subtraction has an error as he should have added the opposite of -262.

- b. At what elevation will Diego start on the second day? Show your work.
6,521 feet

DIFFERENTIATE INSTRUCTION

It may be helpful to provide a visual entry point for each problem where applicable. For example, question 3 describes the number of miles a student ran. Encourage students to represent this problem using a tape diagram to provide a concrete visual, or model this for students who are struggling to conceptualize the problem.

3. Ashlyn was given the following problem to solve.

Fernando runs after school to prepare for the cross-country season. On Monday he ran 4.34 miles, Wednesday he ran 6.56 miles, and on Friday he ran 1.375 miles. His teammate Marcus only ran $\frac{4}{10}$ the number of miles as Fernando. How many miles did Marcus run?

Ashlyn solved the problem with the steps shown.

Ashlyn's Work

$$\begin{array}{r}
 4.34 \\
 +6.56 \\
 \hline
 10.90 \\
 +1.375 \\
 \hline
 12.275 \text{ miles}
 \end{array}$$

Fernando ran 12.275 miles.

- a. Identify what Ashlyn did correctly and incorrectly.

Correct	Incorrect
Ashlyn correctly added Fernando's miles from Monday and Wednesday.	Ashlyn made two mistakes. She incorrectly added 10.90 and 1.375. Also, she should have multiplied by $\frac{4}{10}$ instead of dividing. She also mislabeled her answer as how many miles Fernando ran when she was looking for how many miles Marcus ran.

- b. How many miles did Marcus run? Show your work.
4.91 miles

4. Craig was given the following problem to solve.
- Rhonda makes a specific hair dye color by mixing $1\frac{1}{8}$ ounces (oz) of red tone, 0.75 oz of gray tone, and $\frac{5}{8}$ oz of brown tone. If Rhonda needs to make 20 oz of the same color dye, how much of each color tone is needed?

Craig solved the problem with the steps shown.

Craig's Work

$$1\frac{1}{8} + \frac{5}{8}$$
$$\frac{8}{8} + \frac{5}{8} = \frac{13}{8}$$
$$\frac{13}{8} \div \frac{5}{4}$$

$$\frac{5}{4} + 0.75$$
$$1\frac{1}{4} + 0.75$$
$$1.25 + 0.75$$
$$+ 0.75$$
$$\underline{2.00 \text{ ounces total}}$$

$$20 \div 2 = 10 \text{ ounces needed of each}$$

$$1\frac{1}{8} \cdot 10 = 11\frac{5}{8} \text{ oz red}$$
$$0.75 \cdot 10 = 7.5 \text{ oz gray}$$
$$\frac{5}{8} \cdot 10 = \frac{50}{8} = 6\frac{2}{8} \text{ oz brown}$$

- a. Identify what Craig did correctly and incorrectly.

Correct	Incorrect
<i>Craig understood that he needed to first find the number of ounces in 1 batch of dye before finding out how much of each they need for 20 oz.</i>	<i>Craig made an error in rewriting the mixed number as a fraction before adding in the first step.</i> <i>Craig also made an error simplifying $\frac{13}{8}$.</i>

- b. How much of each color is needed to make 20 oz of dye? Show your work.
- Red Tone - $1\frac{1}{8} \cdot 8 = 9$ ounces*
- Gray Tone - $0.75 \cdot 8 = 6$ ounces*
- Brown Tone - $\frac{5}{8} \cdot 8 = 5$ ounces*

3.7.2 ACTIVITY: ERROR ANALYSIS
(CONTINUED)

ENRICHMENT

What advice would you give Craig to avoid making the same error in the future?

3.7.3 COLLABORATION: FINDING MISSING VALUES

TEACHER MOVES

Consider explaining to students that energy usage is measured in kilowatt-hours and that \$0.12 per kilowatt-hour is the cost for each unit for this utility company. Kilowatt-hour doesn't mean the number of kilowatts being used per hour. It is the power consumption of 1000 watts sustained for one hour. Using a laptop all day is approximately 1 kilowatt-hour of energy. Kilowatt-hour is a measure of the energy consumed.

QUESTIONING

Present the kilowatt-hours used over 4 weeks and the kilowatt-hours generated over 4 weeks. Allow the students to calculate the total kilowatt-hours of each over the four weeks. By incorporating an additional operation of addition with the division and subtraction operations already present, students will strengthen the connection to the objective of using three operations.

ENRICHMENT

Prior to completing question 2, assist students with determining errors and correcting mistakes with their classmates. Provide questions that students can ask each other when comparing (and contrasting) values, such as:

- What values and operation did you use to find the kilowatt-hours?
- What values and operation did you use to find the amount due?
- What steps did you follow to reach your answer?



3.7.3 Collaboration: Finding Missing Values

Many homeowners have solar panels that can generate energy to power their homes. The Better Utility Company charges customers who have solar panels using the following criteria:

- Customers are charged \$0.12 per kilowatt-hour for the energy used that is provided by the utility company.
- The utility company does not charge customers for energy they use from their own solar panel(s).
- For every kilowatt-hour of electricity that customers with solar panels generate but do not use, the utility company gives a price adjustment of $-\$0.025$.

Erin Hill's bill for the month of July is missing some information.

Best Utility Company 123 Ohms Way Big City, ST 12345		Erin Hill 456 Long St. Small Town, ST 98765	
Kilowatt-hours used from Best Utility Company	683.67	Charge	\$82.04
Kilowatt-hours generated by solar panel and not used by customer	164	Credit	$-\$4.10$
		Amount Due	\$ 77.94

1. Work with your partner to fill in the missing information on the bill. Round answers to nearest hundredth.
2. Find another classmate other than your partner and compare your missing values.
 - If the missing values for both are the same, have your classmate initial next to each value in your bill.
 - If the missing values are different, discuss your work to determine what error(s) were made. After correcting any mistakes, initial next to each value in each other's bill.



3.7.4 Wrap-Up: Reflection

1. Complete the table by reflecting on the day’s lesson.

Answers will vary.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	

3.7.4 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Post information about what each image is representing close to each image. This will assist students with what responses are expected —the “hand clapping” image is to elicit a positive aspect about the lesson and the “road signs” image is to elicit a confusing aspect about the lesson.

Alternatively, students can edit the image provided or sketch their own image and include their rationale for why they chose that image.